

OSI 7-Layer Model

1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer.

ANS:

Layer Number	Layer Name	Real-World Example (Device/Protocol)
Layer 7	Application	HTTP, HTTPS, FTP, DNS, Web Browser
Layer 6	Presentation	SSL/TLS Encryption, JPEG, MP3
Layer 5	Session	NetBIOS, RPC, Session Management
Layer 4	Transport	TCP, UDP
Layer 3	Network	Router, IP
Layer 2	Data Link	Switch, MAC Address, Ethernet
Layer 1	Physical	Hub, Cables, Fiber Optic, Signals

2. Match the following networking scenarios to the correct OSI layer.

ANS:

- a) Encrypting data before sending: Presentation Layer (Layer 6)
- b) Routing packets across networks: Network Layer (Layer 3)
- c) Converting data to electrical signals: Physical Layer (Layer 1)
- d) Opening a web page in your browser: Application Layer (Layer 7)
- e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

3. Pick any app you use daily and describe how at least three different OSI layers are involved when you send or receive data in that app.

ANS:

WhatsApp Example

- Application Layer (Layer 7):

WhatsApp allows users to send messages, images, and videos through the application interface.

- Transport Layer (Layer 4):

TCP ensures that messages are delivered reliably and in the correct order.

- Network Layer (Layer 3):

IP addresses are used to route data packets between the sender and WhatsApp servers.

- Data Link Layer (Layer 2):

Wi-Fi or mobile network frames carry the data across the local network.

- Physical Layer (Layer 1):

Electrical, radio, or optical signals transmit the data through cables or wireless networks.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

ANS:

Possible OSI Layers Causing the Problem:

- Application Layer (Layer 7)

If the website service (HTTP/HTTPS) is not working properly, the browser cannot open the webpage even though the server is reachable.

- Presentation Layer (Layer 6)

Problems with SSL/TLS encryption certificates may prevent secure websites from loading.

- Session Layer (Layer 5)

A session establishment problem between the browser and the web server could stop the webpage from opening.

- Reasoning:

Since the IP address responds to ping, the Physical, Data Link, and Network layers are working correctly. The issue is likely occurring at the upper layers responsible for web communication and website services.